

YEAR 10 SCIENCE (GENERAL)
PHYSICAL SCIENCES
ASSIGNMENT - MOTION + NEWTON'S LAWS

NAME: SOLUTIONSMARK: 73

Apply relationships between force, mass and acceleration to predict changes in the motion of objects.

Mark	ND	NW	C	HC	O
Mark Range	0-21	22-32	33-46	47-54	55-73

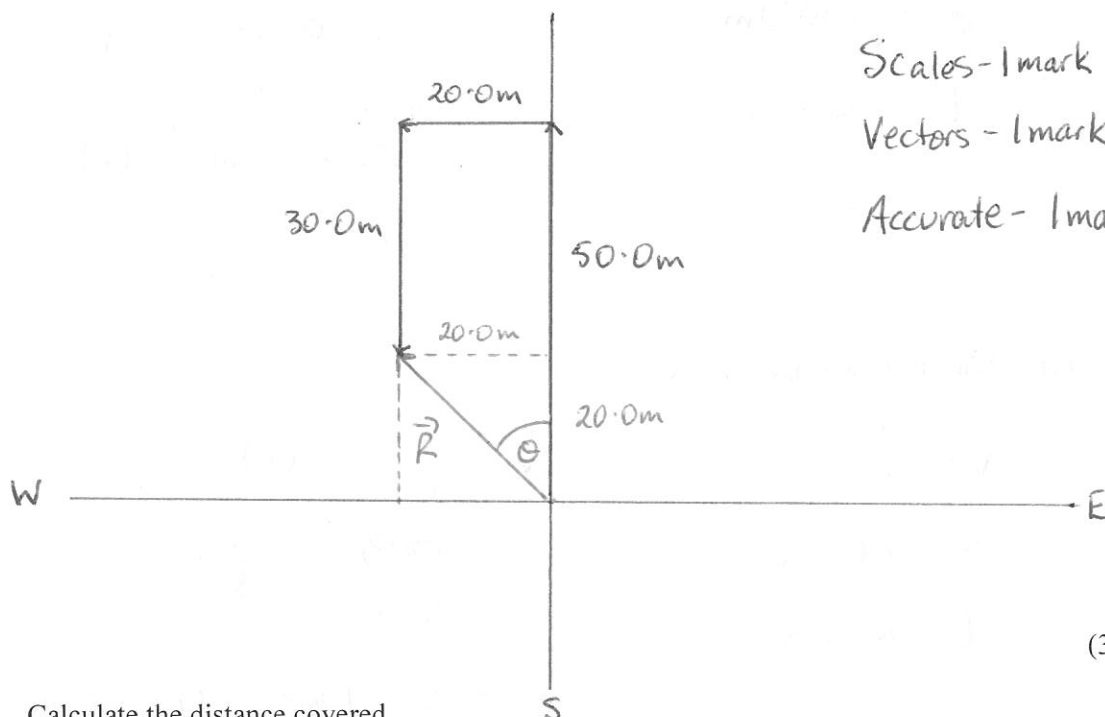
1. Perform the following conversions.

(a) $28.0 \text{ kmh}^{-1} = \underline{7.8} \text{ ms}^{-1}$ (b) $3.7 \text{ tonnes} = \underline{3700} \text{ kg}$ 1 mark each.

(c) $8.3 \text{ minutes} = \underline{498} \text{ s}$ (d) $2.70 \text{ cm} = \underline{0.0270} \text{ m}$ (4)

2. On the school oval, a person runs 50.0 m North, then 20.0 m West and then 30.0 m South. The total time taken for the motion is 40.0 s.

- (a) Draw a vector diagram to show the motion. Draw in the resultant vector (use a different colour). (Use a scale of 1.0 cm = 10.0 m). N



- (b) Calculate the distance covered.

$$d = 50.0 + 20.0 + 30.0 \quad (1)$$

$$= \underline{100.0 \text{ m}} \quad (1)$$

(2)

- (c) Determine the displacement of the person by calculation.

$$\begin{aligned}\vec{R} &= \sqrt{(20.0)^2 + (20.0)^2} \\ &= 28.3 \text{ m.} \quad (1)\end{aligned}$$

$$\begin{aligned}\tan \theta &= \frac{20.0}{20.0} \\ \Rightarrow \theta &= 45.0^\circ. \quad (1)\end{aligned}$$

$$\therefore \underline{\vec{R} = 28.3 \text{ m N } 45.0^\circ \text{ W (or NW)}} \quad (1)$$

(3)

- (d) What is the average speed of the person?

$$\text{speed} = ?$$

$$d = 100.0 \text{ m}$$

$$t = 40.0 \text{ s}$$

$$\begin{aligned}\text{speed} &= \frac{d}{t} \\ &= \frac{100.0}{40.0} \quad (1) \\ &= \underline{2.50 \text{ ms}^{-1}} \quad (1)\end{aligned}$$

(2)

- (e) What is the average velocity?

$$V = ?$$

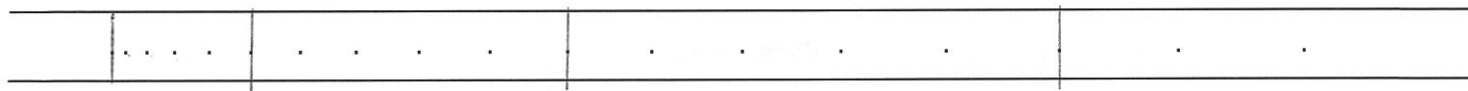
$$s = 28.3 \text{ m}$$

$$t = 40.0 \text{ s}$$

$$\begin{aligned}V &= \frac{s}{t} \\ &= \frac{28.3}{40.0} \quad (1) \\ &= \underline{0.71 \text{ ms}^{-1} \text{ N } 45.0^\circ \text{ W}} \quad (1) \\ &\quad \uparrow \\ &\quad (1)\end{aligned}$$

(3)

3. A group of students investigated the movement of a toy car down an inclined plane in the laboratory, and produced the following ticker-timer tape. The timer produces 50 dots per second.



- (a) Analyse the tape by calculating the velocity for every 5 dot-spacings. Record your values in the table below.

Distance (m)	Time (s)	Velocity (ms^{-1})
0.0	0.0	0.0
0.0185	0.1	0.185
0.042	0.1	0.42
0.065	0.1	0.65

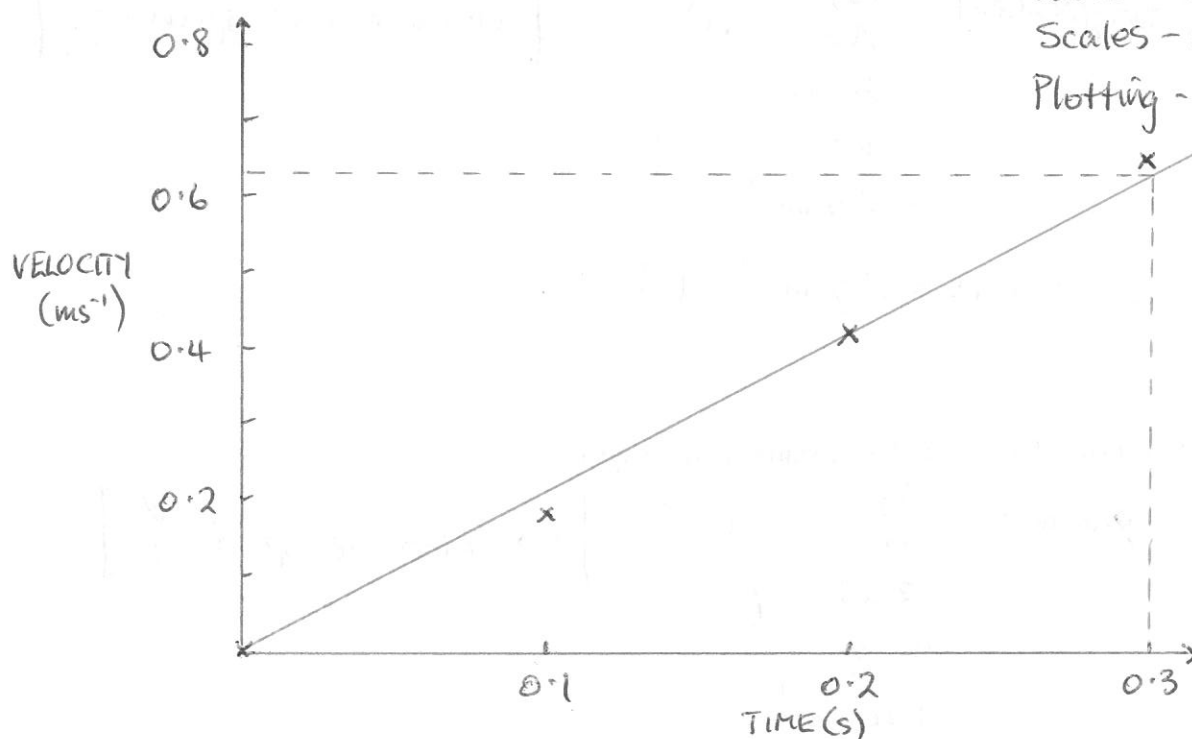
(1)

(1)

(1)

(3)

- (b) Draw a velocity - time graph for these measurements.



Labels + units - 1 mark
Scales - 1 mark
Plotting - 1 mark.

(3)

- (c) Use the graph to determine the acceleration of the toy. Show your working.

$$\text{gradient} = \frac{\Delta y}{\Delta x} \quad (1)$$

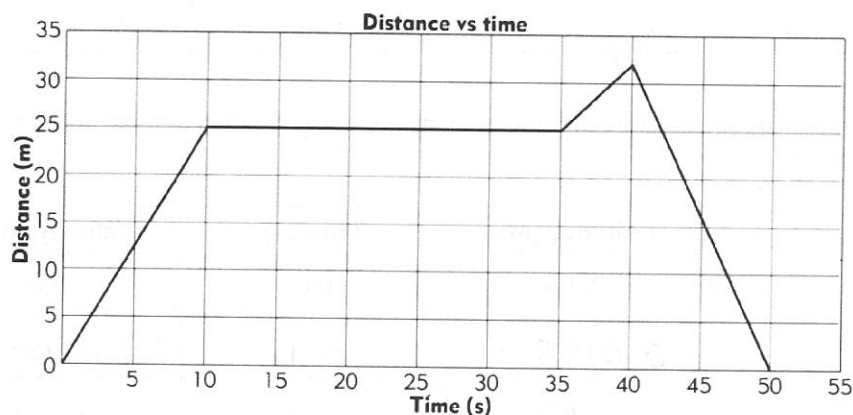
$$= \frac{0.63 - 0}{0.3 - 0} \quad (1)$$

$$= 2.1 \text{ ms}^{-2}$$

$$\therefore \text{acceleration} = 2.1 \text{ ms}^{-2} \quad (1)$$

(3)

4. John decided to take a walk in the playground. He plotted the graph below to show how his distance changed with time during the walk.



- (a) At what time did John stop walking and talk with Brianna? 10 s (1)

- (b) How many seconds did their conversation take? 25 s (1)

- (c) Calculate John's speed during the first 10 s. Show your working.

$$\begin{aligned} \text{gradient} &= \frac{\Delta y}{\Delta x} \quad (1) & \left[\text{Can also use } \text{speed} = \frac{d}{t} \right] \\ &= \frac{25-0}{10-0} \quad (1) \\ &= 2.5 \text{ ms}^{-1} \end{aligned}$$

$$\therefore \text{speed} = 2.5 \text{ ms}^{-1} \quad (1)$$

(3)

- (d) Calculate John's speed after he started moving again.

$$\begin{aligned} \text{gradient} &= \frac{\Delta y}{\Delta x} \quad (1) & \left[\text{Can also use } \text{speed} = \frac{d}{t} \right] \\ &= \frac{32-25}{40-35} \quad (1) \\ &= 1.4 \text{ ms}^{-1} \end{aligned}$$

$$\therefore \text{speed} = 1.4 \text{ ms}^{-1} \quad (1)$$

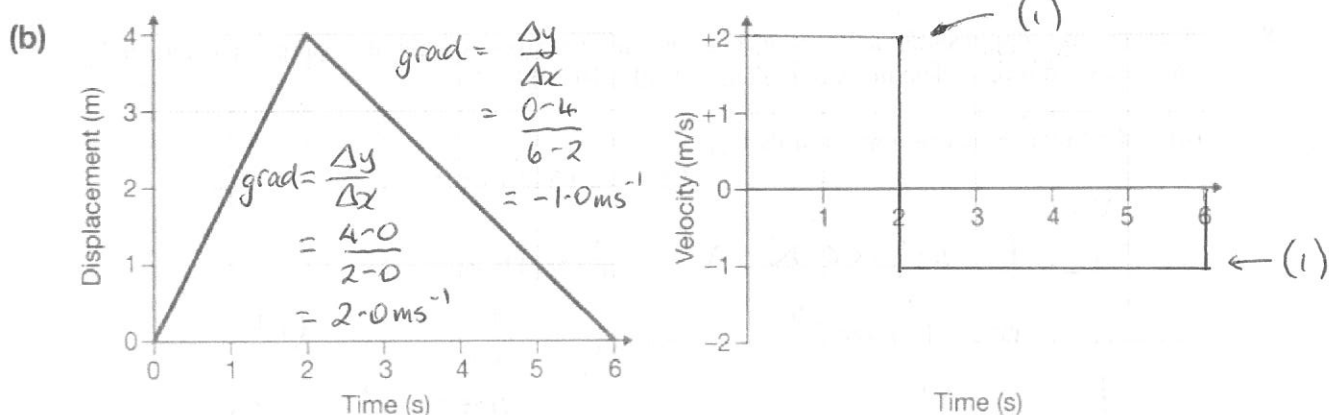
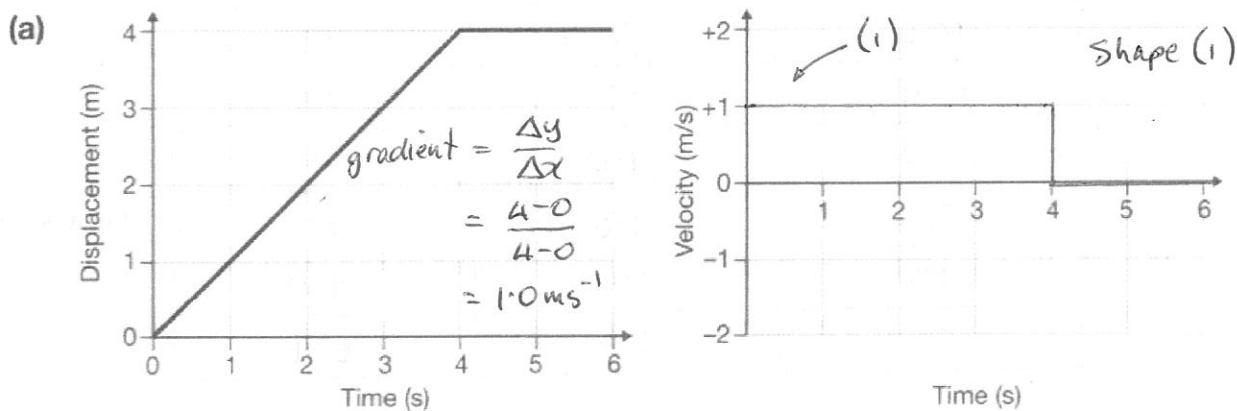
(3)

- (e) What was John's total walking distance?

$$\begin{aligned} d &= 25 + 7 + 32 \quad (1) \\ &= 64 \text{ m} \quad (1) \end{aligned}$$

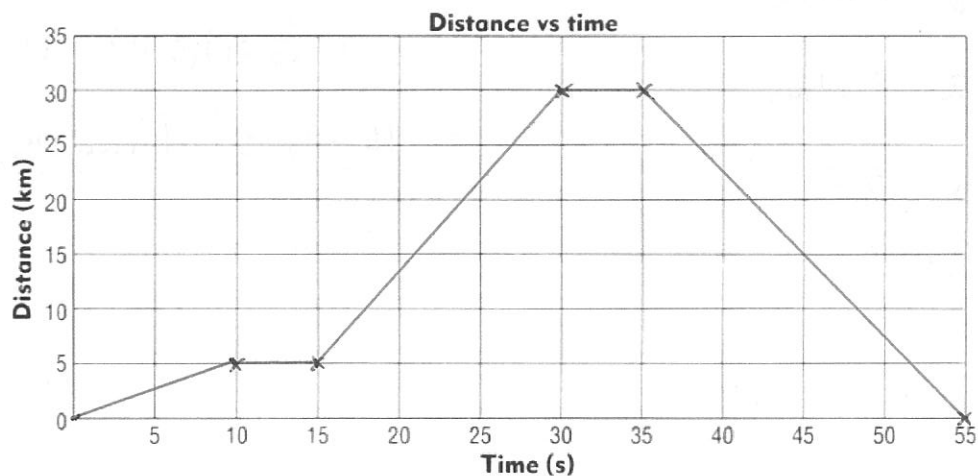
(2)

5. Calculate the gradient for each stage of the two displacement - time graphs below. Use your results to construct the corresponding velocity - time graph for each, using the axes drawn below.



6. Sketch a graph for the following journey on the axes below. (4)

Glenn had to deliver a parcel to Inga. he left home and drove 5 km to fill up with petrol. It took him 10 minutes to reach the service station. Glenn stopped for 5 minutes to fill the car with petrol. He continued on his journey away from his starting point to Inga's house, and at a distance of 30 km he arrived. This part of the journey took another 15 minutes. Glenn talked with Inga for 5 minutes before getting back into his car and returning home. The trip back home took Glenn 20 minutes, during which time he did not stop.



Shape - (1)
 Accuracy - (2).

7. Complete the table below by calculating the missing values.

Force (N)	Mass (kg)	Acceleration (ms^{-2})
20.0	10.0	2.0
18.9	6.5	2.9
56.0	7.0	8.0
24.6	3.6	6.8
108	12.0	9.0
2.9	29.0	0.1

1 mark each.

(6)

8. A car engine supplies $4.50 \times 10^3 \text{ N}$ of force through the front wheels of a car and accelerates it uniformly from rest. The mass of the car is $1.40 \times 10^3 \text{ kg}$.

- (a) Calculate the acceleration of the car.

Forwards is +ve

$$F = 4.50 \times 10^3 \text{ N}$$

$$m = 1.40 \times 10^3$$

$$a = ?$$

$$F = ma$$

$$\Rightarrow a = \frac{F}{m} \quad (1)$$

$$= \frac{4.50 \times 10^3}{1.40 \times 10^3} \quad (1)$$

$$= \underline{3.21 \text{ ms}^{-2} \text{ forwards}} \quad (1)$$

(3)

- (b) Using your answer from part (a), calculate the velocity of the car after 5.0 s.

Take forwards as +ve.

$$v = ?$$

$$u = 0.0 \text{ ms}^{-1}$$

$$a = 3.21 \text{ ms}^{-2}$$

$$t = 5.0$$

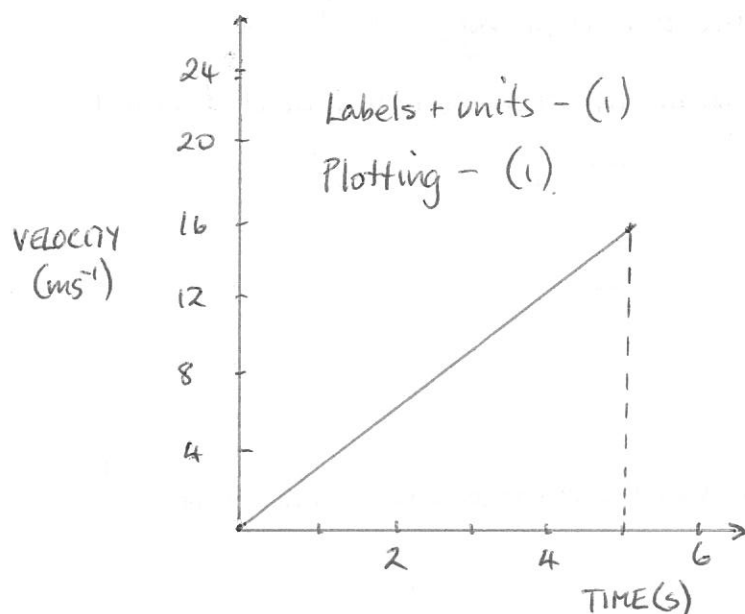
$$v = u + at \quad (1)$$

$$= 0 + (3.21)(5.0) \quad (1)$$

$$= \underline{16.0 \text{ ms}^{-1} \text{ forwards}} \quad (1)$$

(3)

- (c) Draw a velocity - time graph of the motion for 5.0 s and determine the displacement of the car.



$$\begin{aligned} \text{Area} &= \frac{1}{2} b \times h \\ &= \frac{1}{2} (5)(16) \quad (1) \\ &= 40 \text{ m.} \end{aligned}$$

$$\therefore \underline{s = 40 \text{ m forwards.} \quad (1)}$$

(4)

9. A truck of mass 24.0 tonnes is moving at 80.0 kmh^{-1} along Roe Highway when the driver notices a small car that pulled out from a petrol station 60 m ahead. He hit the brakes sharply and just manages to avoid hitting the back of the car. The truck ended up moving at 30.0 kmh^{-1} and took 4.50 s to achieve this.

- (a) What is the deceleration of the truck?

$$v = 30.0 \text{ kmh}^{-1} = 8.33 \text{ ms}^{-1}$$

$$u = 80.0 \text{ kmh}^{-1} = 22.2 \text{ ms}^{-1}$$

$$a = ?$$

$$t = 4.50 \text{ s.}$$

Take forwards as +ve.

$$a = \frac{v - u}{t} \quad (1)$$

$$= \frac{8.33 - 22.2}{4.50} \quad (1)$$

$$= -3.1 \text{ ms}^{-2}$$

$$\therefore \underline{\text{Deceleration} = 3.1 \text{ ms}^{-2} \text{ backwards.} \quad (1)}$$

(3)

- (b) Calculate the average force exerted by the brakes of the truck.

Take forwards as +ve.

$$F = ?$$

$$m = 24.0 \times 10^3 \text{ kg}$$

$$a = -3.1 \text{ ms}^{-2}$$

$$F = ma \quad (1)$$

$$= (24.0 \times 10^3)(-3.1) \quad (1)$$

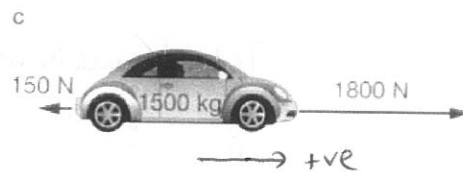
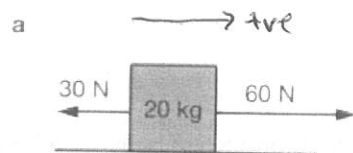
$$= -7.44 \times 10^4 \text{ N}$$

$$\therefore \underline{F = 7.44 \times 10^4 \text{ N backwards.} \quad (1)} \quad (3)$$

10. Explain why you "lurch forward" when standing on a bus or train and the vehicle brakes suddenly.

- You have inertia since you are moving. (1)
- As the vehicle slows, your inertia carries you forwards. (1)

11. Calculate the nett force and the acceleration for the following objects. Show your calculations. (2)



$$\begin{aligned} (a) \quad \Sigma F &= 60 - 30 \\ &= \underline{30 \text{ N to the right}}. (1) \end{aligned}$$

$$\begin{aligned} a &= \frac{\Sigma F}{m} = \frac{30}{20} \\ &= \underline{1.5 \text{ ms}^{-2} \text{ right}}. (1) \end{aligned}$$

$$\begin{aligned} (b) \quad \Sigma F &= 250 - 100 \\ &= \underline{150 \text{ N up}}. (1) \end{aligned}$$

$$\begin{aligned} a &= \frac{\Sigma F}{m} = \frac{150}{75} \\ &= \underline{2.0 \text{ ms}^{-2} \text{ up}}. (1) \end{aligned}$$

$$\begin{aligned} (c) \quad \Sigma F &= 1800 - 150 \\ &= \underline{1650 \text{ N forwards}} (1) \end{aligned}$$

$$\begin{aligned} a &= \frac{\Sigma F}{m} = \frac{1650}{1500} \\ &= \underline{1.1 \text{ ms}^{-2} \text{ forwards}} (1) \end{aligned}$$

(6)